



Research paper

Life Cycle Cost analysis for industrial bioenergy projects: Development of a simulation tool and application to three demand sectors in Africa



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ARTICLE INFO

Article history:

Received 4 July 2021

Received in revised form 11 December 2021

Accepted 2 February 2022

Available online 16 February 2022

Keywords:

Bioenergy

Life Cycle Costing

Levelized Cost of Energy

Combined heat and power

Anaerobic digestion

Combustion

Sub-Saharan Africa

ABSTRACT

Publicly available toolkits for cost calculation in renewable energy projects were reviewed, and their main characteristics, strengths and weaknesses identified. An original Life-Cycle Cost (LCC) modelling toolkit was developed to address some of the identified shortcomings, and to facilitate the calculation of costs of electrical and thermal energy from biomass-based processing pathways. The toolkit was used to compare the Levelized Cost of Energy (LCOE) for electricity and/or heat generation for a 'Bioenergy Case', after the adoption of bioenergy technology, and compared with equivalent costs under a business-as-usual 'Base Case', for selected commercial demand sectors in Sub-Saharan Africa, within a 10 kW to 5 MW scale range. Case Studies of the toolkit's application are described for a smallholder tea factory in Kenya, a wood processing enterprise in Tanzania and an oil palm mill in Ghana, which illustrate its ability to model energy production based on both combustion and anaerobic digestion, under various heat, power and combined heat and power configurations. Sensitivity analysis was carried out to determine the effect on $LCOE_{heat}$ and $LCOE_{electricity}$ of adjustments in key operating parameters, such as biomass feedstock cost, feed-in tariff for export of surplus power to the grid or on-site energy demand. The versatility of the LCC toolkit enables bioenergy projects to be modelled from different perspectives (biomass feedstock diversification, conversion pathways or business models) and for different users (plant managers, project developers or potential investors), and it can be customised and adapted as prices, legislation and technical aspects evolve with time.

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1. Introduction

There exist a number of technology pathways for bioenergy conversion and generation, some of them combustion-based, others based on thermochemical processes such as pyrolysis or gasification, and a third group based on biochemical processes such as anaerobic digestion (AD) or fermentation. Aslani et al. (2018) have analysed these different technologies from a life cycle perspective, concluding that combustion is the most mature option, followed by AD, gasification, thermochemical hydrolysis and pyrolysis; whereas carbonisation, biorefining, biochemical and fermentation technologies are still to reach full maturity, according to the Gartner Hype Cycle for emerging technologies (Gartner Consulting, 2005). The Bioenergy for Sustainable Local Energy Services and Energy Access in Africa - Phase 2' research project (BSEAA2) (NIRAS-LTS et al., 2021) focused on three bioenergy

conversion pathways with a high degree of maturity: combustion for heat production; combustion for combined heat and power (CHP) production with a steam turbine and generator; and AD for biogas production with a reciprocating engine for power generation.

Bioenergy is one of the world's main sources of renewable energy (RE), especially for industrial heat generation applications. Several authors (Ahmadi et al., 2020; Domac et al., 2005; Gan and Smith, 2007; Hillring, 2002) have highlighted its potential to create jobs and foster community development, which is crucial for developing countries such as those in Sub-Saharan Africa (SSA). Modern bioenergy technologies still face challenges to wider deployment, however, highlighting the value of integrated techno-economic analysis to inform investment choices, especially in energy-intensive industrial sectors. Life Cycle Costing (LCC) is a useful approach for evaluating the overall economic performance of a project, and is aligned with Life Cycle Assessment (LCA). A life cycle perspective is relevant when evaluating

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projects that might involve significant upfront costs that are offset by savings at the operational stage, as often seen with bioenergy projects (IRENA, 2015, 2012). The economic performance of an energy project has implications at different stages of its implementation, from inception, installation and commissioning, through to operation, abandonment or end of life. A short-term comparison between different technology options might mean that overspends in one aspect or during one period might not be appropriately balanced by savings enjoyed elsewhere. LCC analysis can holistically compute all relevant costs over a project's lifetime, and includes techniques for successfully allocating costs and levelling them out to generate consistent economic metrics that account for inflation and currency depreciation, facilitating reliable comparison between different projects, technologies and industrial configurations.

A literature review was carried out to identify and understand the state of the art in the analysis of LCC, Levelized Cost of Energy (LCOE) and related sensitivity analysis. Existing LCC/ LCOE tools were also reviewed.

1.1. Life cycle costing (LCC)

Several authors have conducted LCC analysis of bioenergy projects, including both combustion and gasification-based CHP, and AD for biogas production.

For combustion- and gasification-based projects, Ahmadi et al. (2020) systematised a performance analysis to compare the LCOE of four CHP bioenergy pathways based on Internal Combustion Engines (ICE) and Stirling Engines (SE) (chipping–gasification–ICE, pelletising–gasification–ICE, pelletising–combustion–SE, and chipping–fast pyrolysis – ICE), comparing the bio-based scenarios with a baseline scenario reliant on diesel power. Lu and El Hanandeh (2017) performed a combined LCA and LCC analysis of using hardwood thinning residues in six bioenergy conversion scenarios and performed a Monte Carlo-based sensitivity analysis on several of the model input parameters. Liu et al. (2017) carried out economic and environmental modelling of the supply chain of different agricultural feedstocks in the north-eastern USA, basing the costing analysis on the effect of Internal Rate of Return (IRR) and the feedstock-to-energy conversion rate on the Required Selling Price of the feedstock. In their two-part study, Cleary and Caspersen (2015), Cleary et al. (2015) studied the greenhouse gas balance and life cycle impacts of using harvest residues for both small- and large-scale combustion and gasification power plants, proving the environmental benefits of using biomass in industrial facilities and proposing support for small- (rather than large-) scale projects, from an environmental perspective. Lu and El Hanandeh (2017) carried out a techno-economic analysis of small-scale biomass-based CHP under an Energy Services Company (ESCO) model, comparing six different bioenergy pathways with a fossil fuel-based scenario.

Rajendran and Murthy (2019) reviewed techno-economic analyses and LCA of AD projects, concluding that this is a mature technology generating an increasing number of papers and techno-economic and environmental studies. A recent example is the work of Aui et al. (2019), consisting of a combined techno-economic analysis and LCA of an AD plant based on calculation of IRR and greenhouse gas emissions for various scenarios using different feedstocks. Xiao et al. (2020) carried out similar economic and environmental life cycle assessment of biogas production from microalgae. González et al. (2020) performed a combined LCA and LCC of AD using different fruit and vegetable wastes, though did not include a sensitivity analysis.

Other authors (Lee and Den, 2016; Ventura et al., 2013) have carried out a life cycle economic cost analysis of additional bioenergy pathways, including biodiesel, bioethanol and biocharcoal,

concluding that the LCC methodology is well suited to assessing the sustainability and cost of energy products.

According to Muench and Guenther (2013), who carried out a review of LCA of bioenergy projects, the most widely used economic indicators in such analyses are CO₂ mitigation costs, energy provision costs, external costs and profitability. However, an LCC analysis is commonly based on pure economic metrics such as Net Present Value (NPV) or, more usually, the Levelized Cost of Energy (LCOE), which expresses the NPV per discounted unit of energy.

1.2. Sensitivity analysis in cost modelling

Sensitivity analysis is widely applied when studying the economic or environmental performance of bioenergy projects. However, the vast majority of the reviewed studies on LCC of bioenergy projects (Ahmadi et al., 2020; Aui et al., 2019; Cleary and Caspersen, 2015; Cleary et al., 2015; Lee and Den, 2016; Liu et al., 2017; Wood and Rowley, 2011; Xiao et al., 2020) applied this methodology under a *ceteris paribus* approach, i.e. changing one input at a time, rather than performing a multi-variate sensitivity analysis that allows interdependencies between different parameters to be explored. Multi-variate sensitivity analysis is not standard practice, despite its benefits in evaluating the relevance of multi-factorial input parameters (Lamboni et al., 2011).

According to the consulted publications, the most usual practice, almost always in combination with a LCA, is to use techno-economic assessment to estimate the performance, emissions and cost of power plants (Frey and Zhu, 2012), with no specific modelling tool being adopted.

1.3. Review of existing LCC/ LCOE tools

LCOE refers to “the estimates of the revenue required to build and operate a generator over a specified cost recovery period” (US Energy Information Administration (EIA), 2021). According to Hussain et al. (2020); Raikar and Adamson (2020) and Shen et al. (2020), LCOE is a metric that is widely used for comparing lifetime costs between different energy generation pathways or vectors. It is also used for determining the break-even energy selling price over a generation plant's lifetime cost (Agbor et al., 2016). It is therefore a key metric for long-term energy planning (Raikar and Adamson, 2020) and for informing decision-making based on energy costs (Hansen, 2019; Shen et al., 2020). For instance, Ghose and Franchetti (2018) calculated the LCOE of several food waste-to-energy AD projects; Chambon et al. (2020) and Ji et al. (2021) optimised the design of stand-alone hybrid solar photovoltaic (PV) – biomass CHP systems based on LCOE; Oyedun and Kumar (2017) and Xu et al. (2020) used LCOE to compare electricity costs from different co-firing projects; Kost et al. (2018) compared the energy cost ranges of different RE technologies using LCOE; Hussain et al. (2020) modelled and compared three biomass-based Rankine cycle power plants; and IRENA (2012) evaluated a number of bioenergy projects to establish the cost of power generation from biomass.

The LCOE should include capital costs (CAPEX, being the initial investment to build and commission energy infrastructure, and replacement costs projected at a given future year or certain years), operational costs (OPEX, including fixed and variable Operation and Maintenance [O&M] costs) and disposal or abandonment costs (ABEX, although in many cases the residual value might counteract the removal costs). LCOE therefore includes all financial obligations, taxes or insurances, in addition to all direct costs (Dincer and Abu-Rayash, 2020). It is a metric strongly

affected by local financial conditions and the lifetime of the plant (Kost et al., 2018).

A literature review was carried out to characterise existing LCOE and/or LCC tools. This helped ensure that the cost modelling and simulation tool to be developed under the BSEAA2 project would add genuine value to the bioenergy cost assessment tools that are already publicly available. The most important of these are detailed below. All of them are Excel-based spreadsheet tools.

There are two biomass-specific tools:

- Cost of Renewable Energy Spreadsheet Tool (CREST), from the National Renewable Energy Laboratory (NREL) of the US Department of Energy (National Renewable Energy Laboratory (NREL), 2010).
- Biomass-based Energy Cost Calculators from the Department of Biological & Agricultural Engineering, University of California – Davis (UC Davis) (California Biomass Collaborative, 2016).

There are two more general tools covering various RE sources:

- Danish LCOE Calculator from the Danish Energy Agency (Breum, 2016).
- Agora Energiewende LCOE calculation tool (Fürstenwerth, 2014).

One LCOE tool for wind farms assessment was also reviewed:

- MegaVind LCOE tool, from Wind Denmark, a cluster of wind energy companies (MegaVind, 2015).

The CREST uses a cash-flow model for economic assessment of bioenergy projects based on AD. It permits the evaluation of state and federal support systems, and is intended to be used by policymakers, regulators, project developers and investors. The main outputs are net Year One Cost of Energy (COE) and net nominal LCOE, in USD. The tool requires the user to input almost all variables, since it does not incorporate demand estimation or technology costs and feedstock databases. For example, system sizing, system cost and feedstock energy content must be introduced by the user, requiring a significant amount of project development data. Moreover, although the tool offers the possibility of introducing very detailed input data – especially on costs – it is designed only for AD projects, leaving projects based on combustion and other bioenergy pathways out of scope. CREST is usefully designed for modelling a number of RE technologies, for which additional spreadsheet tools are available: solar PV, wind, geothermal and fuel cell.

UC Davis offers an Energy Cost Calculator with six variants: generic biomass power plant, CHP, biomass gasification/power generation, biogas, landfill gas-to-energy and hydrogen (California Biomass Collaborative, 2016). This multi-file tool is based on a cash-flow model that provides a cost analysis for a fixed 20-year planning horizon. Its main outputs are a comprehensive financial analysis leading to Levelized Annual Cost (LAC), in both current and constant USD; and a sensitivity analysis based on: capital cost, fuel cost, heat price, debt ratio, debt interest rate, cost of equity, net station efficiency and capacity factor. The tool can be applied to a range of bioenergy generation pathways, but feedstock characteristics and project costs must be manually introduced, and system sizing must be anticipated in advance.

The other tools identified are not specifically designed for bioenergy projects. For instance, the Danish Energy Agency's LCOE Calculator (Breum, 2016) compares a variety of energy generation pathways, from fossil fuel-based generation plants (coal, natural gas) and nuclear, to large-scale and small-scale PV systems and wind power (on-shore, near-shore and off-shore). The tool includes a number of bioenergy project costs, but all are for large-scale installations (e.g., 80 MW steam turbines).

While the user can manually adjust the cost parameters, they are derived from actual reviewed projects and are therefore not very flexible.

The Agora Energiewende tool (Fürstenwerth, 2014) is a comprehensive LCOE calculation and comparison tool that includes sensitivity analysis and assessment of required Feed-in Tariff (FiT), but only covers combined cycle gas turbines, hard coal, nuclear, wind and solar PV power (at both utility scale and residential rooftop scale), thus does not provide a cost calculation option for bioenergy generation projects. Its sensitivity analysis is nonetheless very comprehensive and permits adjustment of investment cost, weighted average cost of capital, project lifetime, fixed operating cost, fuel cost, CO₂ certificate cost, efficiency and utilisation factor. Lastly, the LCOE tool from MegaVind (MegaVind, 2015) is specifically designed for wind power farms.

Cost breakdown is an important characteristic for a Decision Support System (DSS) to explore the trade-offs between different design parameters. Some of the models can perform sensitivity analyses, which are useful informative mechanism for decision-makers, but do not always provide an optimal sizing range for the adjustable input parameters. In addition, none of the reviewed tools incorporate energy demand estimation per unit of industrial output, a feature that would help users to better size the energy generation plant and estimate the associated costs from the point of view of an industrial process, rather than an energy practitioner. The analysis of available toolkits also reveals a lack of internal databases of feedstock characteristics that would help users to perform cost analyses without the need to manually introduce technical characteristics that might fall outside their own expertise – such as feedstock Lower Heating Value (LHV), specific biogas yield from different feedstocks or biogas methane content.

Table 1 summarises the main features and gaps identified in the existing tools.

Lastly, as the rationale behind such tools is to permit assessment of a wide range of potential projects and industrial plant configurations, they will be more useful as more bioenergy pathways are included. There is therefore a need to include pathways based on both combustion (and related CHP) and AD.

The analysis further reveals that the following features are not fully covered by any single tool:

- Comprehensive sensitivity analyses, including multivariate analysis, trend studies and measures of dispersion.
- System pre-sizing check according to energy demand.
- Energy demand estimation, depending on intended applications.
- Feedstock characteristics database including (e.g.) LHV, methane content, biogas yield and CO₂ impact.
- Characteristics of the base case energy carrier (i.e., natural gas, diesel or another feedstock): LHV, CO₂ impact.
- Conversion technology cost estimation (based on an internal database of default values).
- Inclusion of both AD and combustion-based bioenergy pathways, including CHP in the latter.
- Estimation of target subsidy and/or tariffs to achieve economic viability (e.g., to break even the LCOE).
- Readability and ease of interpretation of results, through use of comparative plots.

In addition, given the scale range of interest for the BSEAA2 project, the desired tool for this purpose should be customisable for a power production range between 10 kW and 5 MW.

The remainder of this paper is structured as follows:

- Section 2 summarises the Methodology of the LCC analysis and related metrics.
- Section 3 details the Results from three case studies.
- Section 4 summarises the main Conclusions from the work.

Table 1
Features and gaps of reviewed LCOE tools.

Tool	Main features	Identified gaps
CREST	• Net COE and LCOE calculation • Cash-flow based (detailed) • AD is considered	• Lacks internal database so almost all input data must be provided • No system sizing calculation • Does not cover combustion and CHP • No sensitivity analysis
UC Davis Energy Cost Calculators	• LAC calculation • Cash-flow based (simplified) • Cover up to 6 different types of projects • Sensitivity analysis <i>ceteris paribus</i> is included	• No system sizing calculation • No detailed cash-flow analysis • No multi-variate sensitivity analysis
Danish LCOE calculator	• LCOE calculation by concept (investment, fuel, etc.) • Cash-flow (simplified) • Can analyse wide range of energy pathways, including nuclear and solar • Comprehensive database of projects' costs. For biomass, large-scale projects data included	• Does not cover small-scale biomass projects • Lacks flexibility and ease of data input • No detailed cash-flow analysis • No sensitivity analysis
Agora	• LCOE calculation and comparison • Required FIT calculation • Covers RE projects and comparison with fossil-fuel alternatives • Sensitivity analysis <i>ceteris paribus</i> included	• Does not cover bioenergy projects • No cash-flow analysis • Results not easy to read and interpret • No multi-variate sensitivity analysis
MegaVind LCOE tool	• LCOE calculation • Cash-flow based (detailed) • Wind farm assessment • Thorough financial analysis	• Does not cover bioenergy projects • No sensitivity analysis • Lacks easy-to-read results (plots)

2. Methodology

The methodological approach comprises the general LCC framework approach – and LCOE calculation – and the evaluation of externalities included in the general cost modelling methodology.

2.1. General LCC framework

LCC analysis, also known as total cost accounting, is an economic evaluation process that determines the full costs of a building, structure, system or complete component over its useful life, while defining alternative investment options in buildings or operating activities. By using the LCC of a biomass system, a baseline case (or a Business-as-Usual scenario, BAU) can be compared with an alternative 'Bioenergy Case'.

LCC analysis design and practice is standardised in ISO 15686-5 (ISO TC 59/ SC 14, 2017), which establishes the necessary criteria to analyse Life Cycle Costing. To be compliant with this standard, the same basic structure for analysing costs of bioenergy alternatives has been adopted in the tool, though simplifying some of the initial hypotheses (see Fig. 1):

Based on this framework for LCC analysis, LCOE is a useful metric for comparing BAU energy solutions with a proposed bioenergy alternative (the Bioenergy Case). LCOE is commonly used for comparing energy generation systems (Dincer and Abu-Rayash, 2020; Hansen, 2019; Hussain et al., 2020; Raikar and Adamson, 2020; Shen et al., 2020; Xu et al., 2020) because it helps to assess and compare different methods of energy production. Since the BSEAA2 project addressed the generation of electricity as well as heat, the model specifies $LCOE_{\text{heat}}$ for thermal energy and $LCOE_{\text{electricity}}$ for electricity. In order to clarify the cost build-up, the LCOE is split into CAPEX, OPEX and ABEX.

The LCC modelling approach is summarised in Fig. 2. The most relevant input parameters are highlighted in red (if they are technology or Demand Sector-dependent) or in blue (if they are country-dependent).

Adopting this approach, the LCC is modelled as shown in Eq. (1):

$$LCC [\$/] = CAPEX \left[\frac{\$}{\text{kW}} \right] \cdot P_n [\text{kW}]$$

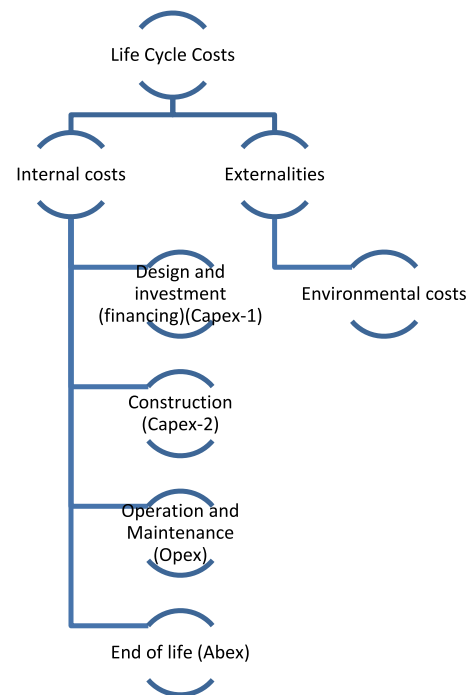


Fig. 1. Life Cycle Costs structure.
Source: Authors.

$$+ \sum_i \frac{\left(OPEX \left[\frac{\$}{\text{kWh}} \right] \cdot E_{\text{prod}} \left[\frac{\text{kWh}}{\text{year}} \right] + OPEX \left[\frac{\$}{\text{year}} \right] \right) \cdot \text{years} + ABEX \left[\frac{\$}{\text{kW}} \right]}{(1 + DR)^i} \quad (1)$$

Where $CAPEX \left[\frac{\$}{\text{kW}} \right]$ refers to initial investment (product acquisition + installation and commissioning) in USD (\$) per kilo-Watt (kW); $OPEX \left[\frac{\$}{\text{kWh}} \right]$ refers to the variable costs of operation, i.e. those dependent on the energy production; $OPEX \left[\frac{\$}{\text{year}} \right]$ refers to the fixed costs of operation per year; $ABEX \left[\frac{\$}{\text{kW}} \right]$ refer to

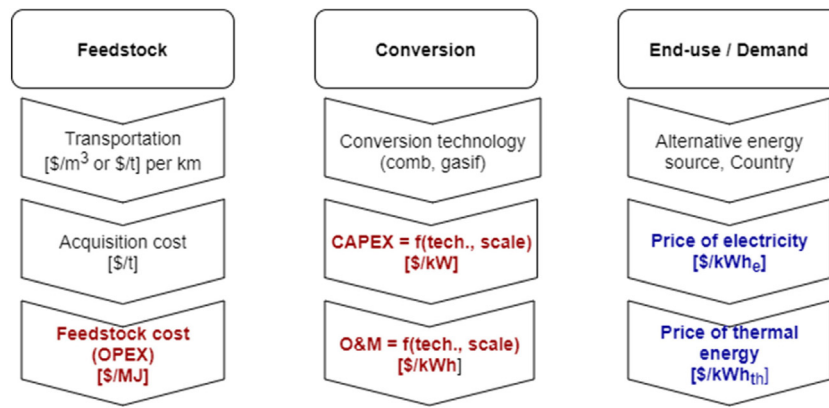


Fig. 2. Relevant pathway costs. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)
Source: Authors.

abandonment expenditures, including residual value, if any, as income; and DR is the Discount Rate. The ABEX is estimated as a standard percentage of CAPEX according to (Larsson, 2012), while all other costs are case-specific.

The LCOE is calculated as shown in Eq. (2):

$$LCOE [\$/MWh] = \frac{LCC[\$]}{E_{prod} \left[\frac{MWh}{year} \right] \cdot years / (1 + DR)^i} \quad (2)$$

Where $LCC[\$]$ refers to the LCC as calculated in Eq. (1); $E_{prod} \left[\frac{MWh}{year} \right]$ refers to the annual energy production in MWh; and DR is the Discount Rate. In CHP-based bioenergy projects two different CAPEX and fixed OPEX allocation methods are integrated into the LCC toolkit:

- Ideally, if the costs breakdown between (i) the additional investment needed by the dedicated biomass electricity power plant and (ii) by the heat production unit is available, the end-user is encouraged to manually allocate them to $LCOE_{electricity}$ and $LCOE_{heat}$, respectively.
- Alternatively, if the end-user can only estimate the overall CHP system CAPEX, the LCC toolkit also offers the possibility to allocate CAPEX and fixed OPEX to $LCOE_{heat}$ and $LCOE_{electricity}$ considering a cost share proportional to the CHP unit output and plant consumption.

2.2. Consideration of externalities

The adoption of biomass energy technologies is also dependent on the general framework of inclusion of externalities. Externalities occur in situations when the producer is not the only one paying for the cost of production. The most usual externalities in biomass transformation arise from climate change effects (due to uncaptured CO_2 emissions) and to the high geopolitical costs of fossil fuel exploitation. For the BSEAA2 project, externalities include the necessary treatment of certain Chemical Oxygen Demand (COD)-intensive effluents (e.g., in the palm oil and dairy industries). Such externalities are impacts that are included in the tool, in order to be economically efficient (using the Coase polluter-pays principle (Coase, 1987, 1960)).

The CO_2 externality effect is handled by including a variable to incorporate the value of CO_2 in the global market, as shown in Eq. (3). Continuous fluctuations in the value of CO_2 reduce the trustworthiness of the tool and for the analysed cases, a value of zero was applied. In order to keep providing insight on such a relevant parameter, the tool calculates the annualised cost of a tonne of avoided CO_2 emission. If the obtained value is negative, then the investment does not require CO_2 internalisation to be

financially viable, whereas if the obtained value is positive, it indicates that the necessary spot price per tonne of CO_2 has been achieved to make the investment financially attractive.

$$\text{Annualised } CO_2 \text{ cost } [\$ / tCO_2] = \frac{LCC[\$]}{CO_{2emissions} \left[\frac{tCO_2}{year} \right] \cdot years / (1 + DR)^i} \quad (3)$$

Where $LCC[\$]$ refers to the LCC as calculated in Eq. (1), $CO_{2emissions} \left[\frac{tCO_2}{year} \right]$ refers to annual CO_2 emissions in tCO_2 , and DR is the Discount Rate.

In the case of waste effluents requiring treatment, the externality accounted for is the cost per tonne or cubic metre of waste to be treated ($Waste_{cost} [\$ / tonne]$ or $[\$/m^3]$). In most countries, legislation exists for the treatment of effluents with high Biological Oxygen Demand (BOD) and COD, as typically found in palm oil mill effluent, dairy wastewater, sisal processing residues and the oily residues from extracting avocado oil.

3. Results and discussion

The methodological approach described in Section 2 was followed to design, build and test an Excel-based LCC modelling toolkit. This toolkit was validated against three case studies representing researched bioenergy generation pathways in SSA: (i) Combustion for direct thermal application in a smallholder tea factory in Kenya; (ii) CHP from combustion in a wood processing company in Tanzania; and (iii) AD from palm oil processing in a large-scale mill/refinery complex in Ghana.

For all three case studies, the period chosen for the economic analysis is 10 years, which is a standardised period for economic analysis. Longer periods are generally applied in sustainability assessments and shorter periods are considered by investors, neither being necessarily indicative of the functional lifetime of a particular project.

3.1. LCC toolkit development

The objective of the economic competitiveness analysis was to use the LCC modelling toolkit developed by the authors to compare the energy costs under the Base Case and the Bioenergy Case, on the basis of a 10-year discounted cash flow analysis, and to investigate the main economic drivers for the wider adoption of bioenergy in the given demand sector.

The following features of the toolkit are worth noting:

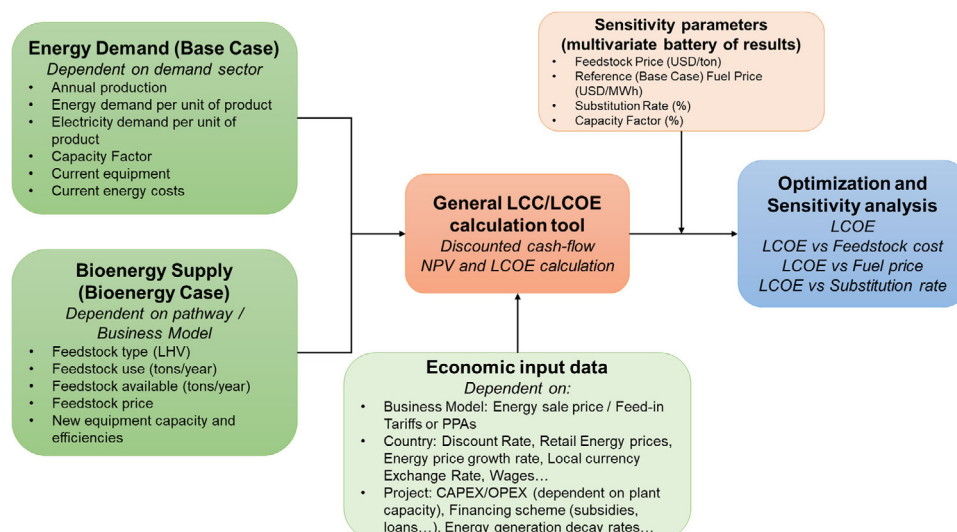


Fig. 3. General layout and cost calculation flow in the LCC modelling toolkit.

Source: Authors

- Energy generation is modelled under a BAU ('Base Case') scenario and a Bioenergy Case scenario, for a number of industrial demand sectors.
- The main economic indicator considered is the LCOE, in \$/MWh. LCOE comprises CAPEX (upfront investment and other amortisable costs), OPEX (personnel, consumables and operating costs such as fuel costs) and ABEX (abandonment expenditures). The model allows LCOEs to be calculated for both electricity and heat, enabling its use in different types of bioenergy projects (combustion for heat generation, CHP, electricity generation only).
- The model allows sensitivity analyses to be carried out on the LCOE, considering a range of values for relevant input parameters tailored to each Demand Sector.
- The toolkit permits multi-variate sensitivity analysis, to determine the influence of a wide range of input parameters on the LCOE.
- The toolkit has a comprehensive internal database covering various economic input data, including: technology type and equipment cost, O&M cost estimates and fuel characteristics (LHV, CO₂ emissions factor). Such values provide the user with default estimated values, which can be manually overridden to apply case-specific data, if available.

The structure of the LCC modelling toolkit is represented in Fig. 3.

The tool includes a number of pre-defined values from an internal database of country-specific and technology-specific sources for data, technology costs and feedstock and biogas characteristics. The user needs only to input any additional parameters that are project-specific. The cost modelling is based on the parameters shown in Fig. 4. Efficiencies of energy generation equipment were estimated according to (Gonzalez et al., 2014) and project site visits. The tool allows the user to select the period of the economic analysis, from a single year up to 20 years. It has two input data sheets and one results sheet. The user must supply project-specific input data and can overwrite default values from the internal database.

The tool also has three cash-flow spreadsheets used as the core calculation engine, an internal database sheet and a sensitivity analysis sheet, in which the results for each simulation are generated. Given that it performs a multivariate sensitivity analysis, the number of simulations might run from dozens up to hundreds

or thousands, and the user is automatically warned if the time required to perform the simulation might be very long.

3.2. Case study 1: Combustion: Smallholder tea factory in Kenya

Kenya is the world's largest exporter of black tea, earning over \$1.4 billion in foreign exchange from tea exports in 2018. Over 60% of Kenya's tea comes from 650,000 small-scale growers, whose tea is processed at 70 independent factories under 54 farmer-owned companies. These companies are, in turn, shareholders in Kenya Tea Development Agency (KTDA) Holdings Ltd. (IFC, 2014). The balance of Kenya's tea comes from privately owned estates.

The processing of black tea is highly energy intensive and requires thermal energy for withering the freshly picked leaves and for drying the leaves after fermentation. At KTDA factories, the heat is derived from fuelwood, but some factories have been trialling the use of alternative biomass-based fuels, such as briquettes made from densified bagasse or sawdust.

Table 2 summarises the data used in the LCC model for KTDA's Makomboki Tea Factory in Murang'a County, which has been substituting a portion of its fuelwood with bagasse briquettes (as the Bioenergy Case), compared with a Base Case scenario of generating heat using fuelwood only, which is the industry standard. The Bioenergy Case assumes that 8% of the required thermal energy comes from briquettes, which reflects the reality at Makomboki for the 3-year period 2018–2020.

Applying these input parameters, the LCC model shows that the Makomboki tea factory has lower LCOE for heat under the Base Case than the Bioenergy Case (Fig. 5), being \$50.84 per MWh_{th} against \$54.54 per MWh_{th}. In other words, the cost of thermal energy is higher as a result of the addition of briquettes to the factory's fuel supply. The 8% substitution of fuelwood by briquettes results in a 7.4% increase in the cost of thermal energy, per tonne of tea processed. This cost differential is likely to be one reason that no more than three KTDA factories are understood to have purchased briquettes.

The economic viability of the partial substitution of fuelwood by biomass briquettes is influenced by a number of factors, whose impact on LCOE_{heat} can be estimated through sensitivity analysis. In the charts below, the linear regression lines are a best-fit for the results from hundreds of simulated scenarios.

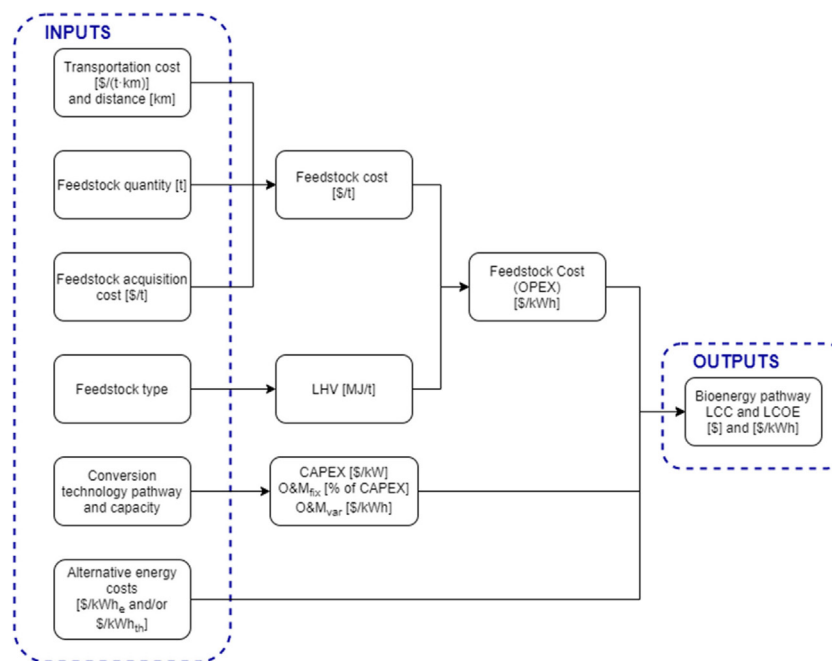


Fig. 4. Summary of bioenergy pathway cost modelling toolkit inputs – outputs scheme.
Source: Authors.

Table 2
Key economic modelling parameters for Case Study 1.

Category	Parameter	Value	Reference
General parameters	Discount rate	10%	(Central Intelligence Agency, 2017)
	General price growth rate	8% (Consumer Price Index)	
	Energy price growth rate	8% (Energy Price Index)	
	Electricity price	\$152.44/MWh (Kenya Power Small Commercial tariff)	(EPRA, 2012)
	Exchange rate	102.33 KES/\$ (3 yr average)	–
Base Case	Energy intensity	22.7 GJ/t made tea	(Inoti, 2016; Kimari, 2012)
	Tea output	6,252 t/year	(EED Advisory, 2018)
	Fuel used	Fuelwood. LHV 13 GJ/t @ 25% MC	Field Visit, (NIRAS-LTS et al., 2021)
	Fuel cost	80% locally supplied @ \$63/t (\$62 + \$1 for handling and preparation). 20% from Kakuzi ^a or similar @ \$79/t (\$64 + \$15 for transport). Fuelwood prices levelised to 25% MC.	
Bioenergy Case	Biomass substitution rate	8% briquettes (plus 74% fuelwood from farmers; 18% fuelwood from Kakuzi)	Field Visit, (NIRAS-LTS et al., 2021)
	Feedstock used	Bagasse briquettes: LHV 18 GJ/t @ 10% MC. Fuelwood properties as per Base Case.	
	Feedstock cost	Bagasse briquettes: KES 15,660 (\$153)/t plus \$15/t for transport = \$168/t. Fuelwood costs as per Base Case.	

^aKakuzi PLC is a Kenyan company engaging in commercial forestry and the cultivation, processing and marketing of avocados, blueberries, macadamia, tea and livestock. <https://www.kakuzi.co.ke/about-us>.

Fig. 6 shows the sensitivity of LCOE to a range of briquette costs from \$50 to \$175/t (versus a current cost of around \$168/t), under a number of operational scenarios. The tipping point of viability is achieved at a briquette cost of around \$75/t, including VAT and delivery costs, which is substantially lower than the current cost.

Economic analysis shows greater heat costs for the partial substitution of fuelwood by briquettes, giving increasingly poor results as the substitution rate rises. The break-even cost of thermal energy is achieved with a delivered briquette cost of \$75/t, substantially lower than the current cost of \$168/t. There may be boiler performance improvements attributable to the use of

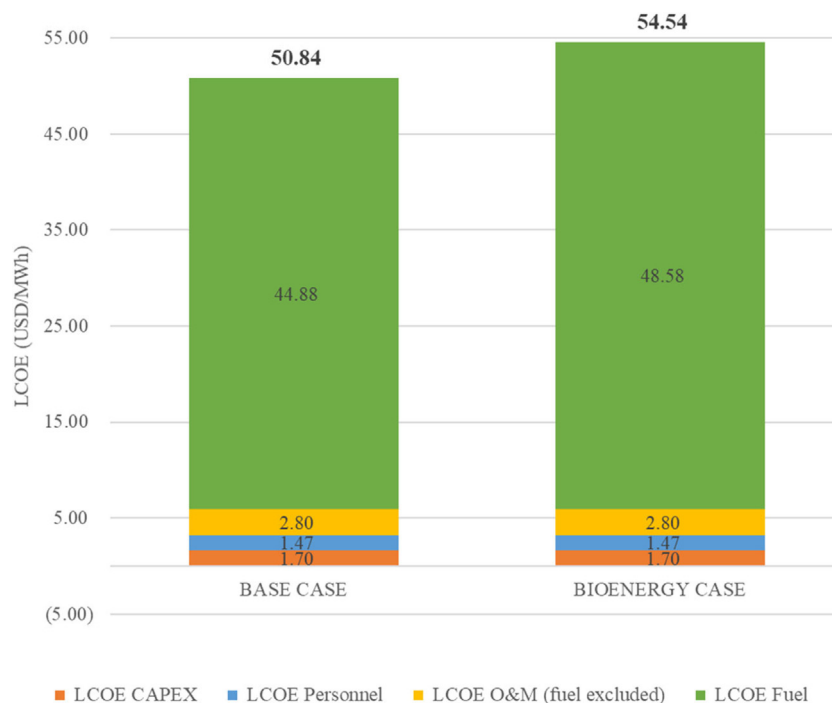


Fig. 5. LCOE comparison for Base Case vs. Bioenergy Case, Makomboki tea factory.

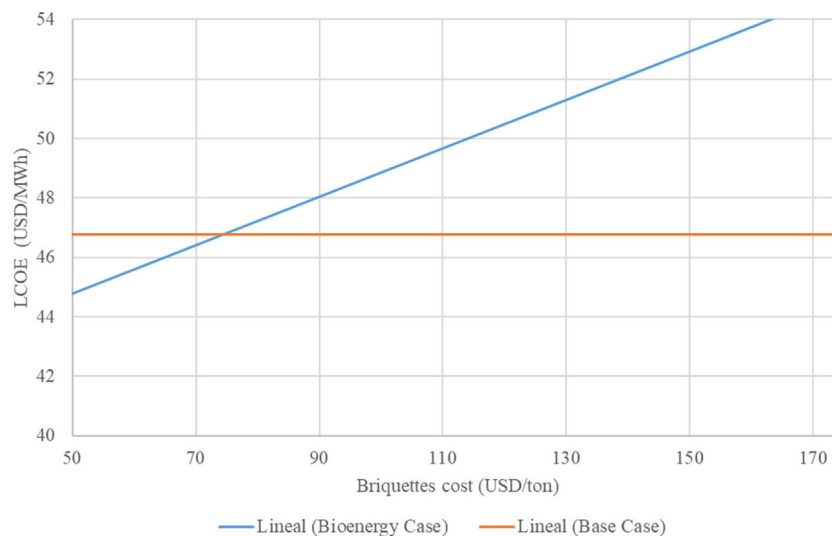


Fig. 6. Sensitivity of $LCOE_{heat}$ to briquette cost.

drier and more standardised briquettes, as a partial replacement for fuelwood, but it seems unlikely that these will be sufficient to outweigh the significant energy cost differential.

Although energy from briquettes is around twice as expensive as from fuelwood, there are commercial benefits that might motivate tea factories to consider using briquettes as supplementary fuel. Such motivating factors might include consumer expectations in ethical markets, the need to ensure that fuel supply is diverse, secure and stable, and the part that innovation can play in building functional relationships with supportive development partners. However, these benefits are currently insufficient to overcome the fuel cost barrier. Despite the presence of ample feedstock and sufficient production capacity amongst KTDA's pre-qualified briquette suppliers, there has been only limited uptake

of briquettes across the factory network. Unless there is a significant change in relative fuel costs, this suggests very limited replication potential within KTDA factories, who can access sustainably produced fuelwood from local farmers (often their own shareholders), KTDA- or factory-owned plantations and external suppliers, at lower cost.

3.3. Case study 2: Combined heat and power: Wood processing enterprise in Tanzania

Tanzania has some of the most extensive forest resources in East Africa, with around 47 million ha (half of the country's land area) under forest, woodland and bush land (FAO, 2021, 2015). Approximately 70% of this area is in productive use, while the rest

Table 3
Key economic modelling parameters for Case Study 2.

Category	Parameter	Value	Reference
General parameters	Discount rate	7% (CIA, 2020)	(Central Intelligence Agency, 2017) (EWURA, 2016) –
	General price growth rate	5.3% p.a. (Consumer Price Index)	
	Energy price growth rate	0% p.a.	
	Electricity price	TZS 157,000 (\$68.48) per MWh (TANESCO T3-MV tariff)	
	Exchange rate	2,293 TZS/\$(3-yr average)	
Base Case	CAPEX	\$4.9 M for low pressure boiler + civil works (~\$472/kW _{th})	Field visit, (NIRAS-LTS et al., 2021)
	Wood chip cost	Zero (internally generated)	
	Wood chip LHV	15.4 GJ/t (wattle logs @ 20% MC)	
	Nominal power	10 MW _{th} (1 x low pressure boiler)	
	Genset for backup power	5% of electricity demand; 187,714 l of diesel per year	
	Capacity factor	80%	
Bioenergy Case	CAPEX	\$10.7 M for 2 high pressure steam boilers + turbine + generator + civil works (\$463/kW _{th} + \$674/kW _e)	Field visit, (Gonzalez et al., 2014; NIRAS-LTS et al., 2021) Field visit, (NIRAS-LTS et al., 2021)
	Nominal power	20 MW _{th} + 2 MW _e	
	Annual energy demand	56,000 MWh _{th} + 10,512 MWh _e	
	Electricity sales to grid	25% of power sold @ FiT of \$86.4/MWh	

is protected. Plantations cover only about 280,000 ha, but provide the majority of the country's industrial wood for construction and furniture, poles for housing and electricity distribution, and feedstock for pulp and paper (FAO, 2015). The main plantation species are pine (~65%) and eucalyptus (~20%) (Unique Forestry & Landuse, 2017). Approximately 80% of Tanzania's forest plantations are located in the Southern Highlands, with the largest at Sao Hill covering around 50,000 ha. This area is home to most of the country's important forest sector companies, which are some of the largest and most successful plantation management and wood processing concerns in the region.

Many forest plantations have low yields and produce poor quality wood, but the opportunities to improve yields and diversify products and services are increasingly recognised (Unique Forestry & Landuse, 2017). One option is to develop CHP production based on wood processing residues. CHP commenced at Mufindi Paper Mills in 1986, followed by the Tanganyika Wattle Company (TANWAT) in 1995. Two other large wood processing enterprises, Kilombero Valley Teak Company and Sao Hill Industries, use sawmill wastes to generate heat for wood drying. Kisiwa Farm on Mafia Island in the Indian Ocean, is developing a small CHP plant to be fuelled with material from coconut plantations under rehabilitation, as well as sawmill residues, to generate electricity through gasification of wood chips.

If not used for energy, residues from wood processing in Tanzania are usually given away or sold cheaply as fuel, or are dumped or burned. Productive use of the biomass for heat and electricity production therefore brings environmental benefits in terms of reduced greenhouse gas emissions (through avoided burning or disposal, avoided use of diesel generators during periods of grid instability or in areas where grid is not available) and more environment-friendly handling of wastes.

The BSEAA2 research explored the landscape for bioenergy use in Tanzania's wood processing industry, to identify the potential for adoption of CHP within the sector, and the opportunities and barriers for wider adoption. The analysis was grounded in the experiences of TANWAT, which was the flagship venture representing the Bioenergy Case for this demand sector.

Table 3 summarises the data used in the LCC model for the TANWAT CHP plant, as the Bioenergy Case, comparing it with a Base Case scenario generating heat-only and drawing its electricity from the grid.

In terms of fixed costs allocation, i.e., CAPEX and fixed O&M costs, this case study considers that the CHP unit fixed costs as well as the costs of the high-pressure steam boiler required for

its operation belong exclusively to LCOE_{electricity}, since the plant was already self-sufficient in heat supply and all the additional costs were incurred to avoid the use of electricity from the grid and backup gensets.

Using these input parameters, Fig. 7 shows the LCOE_{heat} and LCOE_{electricity} in the Bioenergy Case compared with the Base Case. The LCC model shows that TANWAT's CHP plant delivers a significant cost reduction for thermal energy and modest cost reduction for electrical energy. TANWAT is estimated to have achieved a reduction in LCOE_{heat} from around \$23.5 to \$19.0 per MWh_{th}, and a small reduction in LCOE_{electricity} from \$87.7 to \$87.6 per MWh_e.

The value of the annualised CO₂ cost is \$6.85/t CO₂, which means the Bioenergy Case saves 845 t of CO₂ per US\$ million of investment, and a total savings of 4,875 t of CO₂ over the 10-year timeframe of analysis.

Fig. 8 takes the analysis further by offsetting electricity costs with the potential revenues from sale of surplus electricity from the CHP plant to TANESCO, the national power utility, under a Power Purchase Agreement (PPA). In this case, OPEX would be almost fully offset in the Bioenergy Case, and an overall 25% reduction in the net cost of electricity supply would be achieved. The extra revenue from electricity sales to TANESCO is therefore an important part of the economic case for TANWAT's investment in CHP, on top of its savings on the cost of providing heat.

Adjustments were made to selected parameters in the LCC model to determine their impact on LCOE, and the implications of these changing costs for the replication potential of the Bioenergy Case. For this demand sector, the chosen parameters were the plant capacity factor, the TANESCO FiT and the power plant size.

The sensitivity analysis of the 'net' cost of electricity supply (defined as the LCOE_{electricity} minus revenue from sales to TANESCO) is shown in Fig. 9, for a range of potential FiT values from \$60–100 per MWh_e (versus the current FiT for a plant of this size of \$86.4/MWh_e). Across this range of FiTs, the net cost of electricity supply is consistently lower than the Base Case, even for FiTs as low as \$60/MWh_e. The ability to sell surplus electricity into the grid would therefore still be an important enabler of the Bioenergy Case, even at lower FiT values.

The results show a significant cost saving for heat arising from the installation of the CHP system, along with a modest saving for electricity, and the outcome is further improved if revenue from sales of surplus electricity to the grid is included. The positive economic case arises due to TANWAT's uniquely diverse production lines for timber, poles and tannin that result in high demand for process heat, which underpins the economic case for

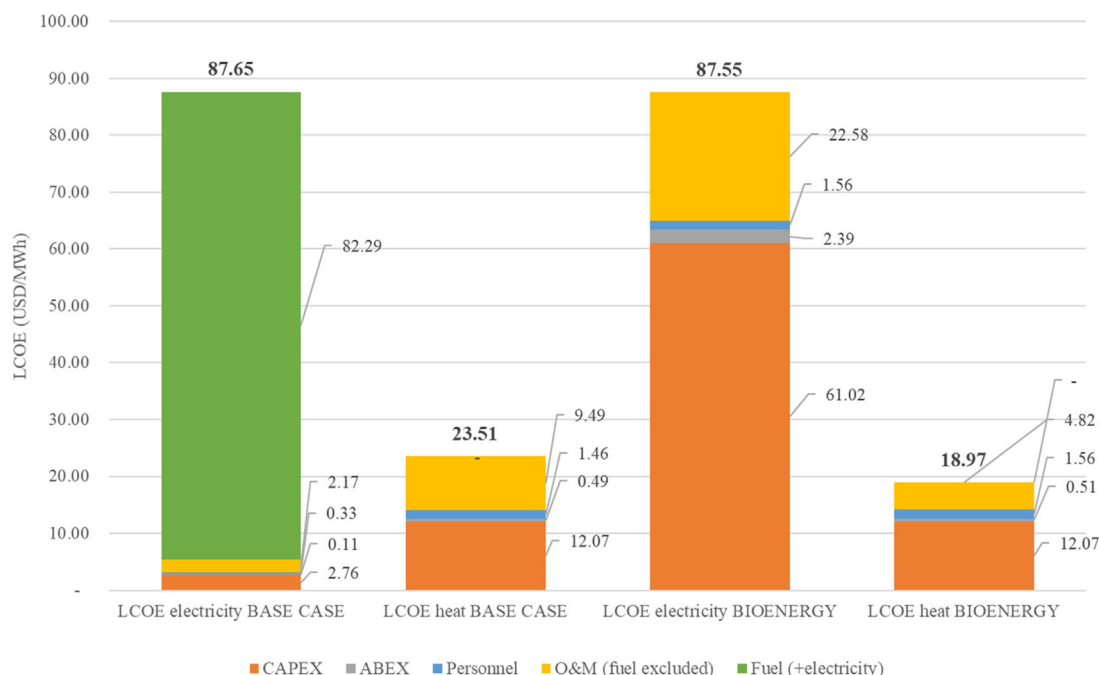


Fig. 7. LCOE comparison for heat and power at TANWAT, Base Case vs. Bioenergy Case.

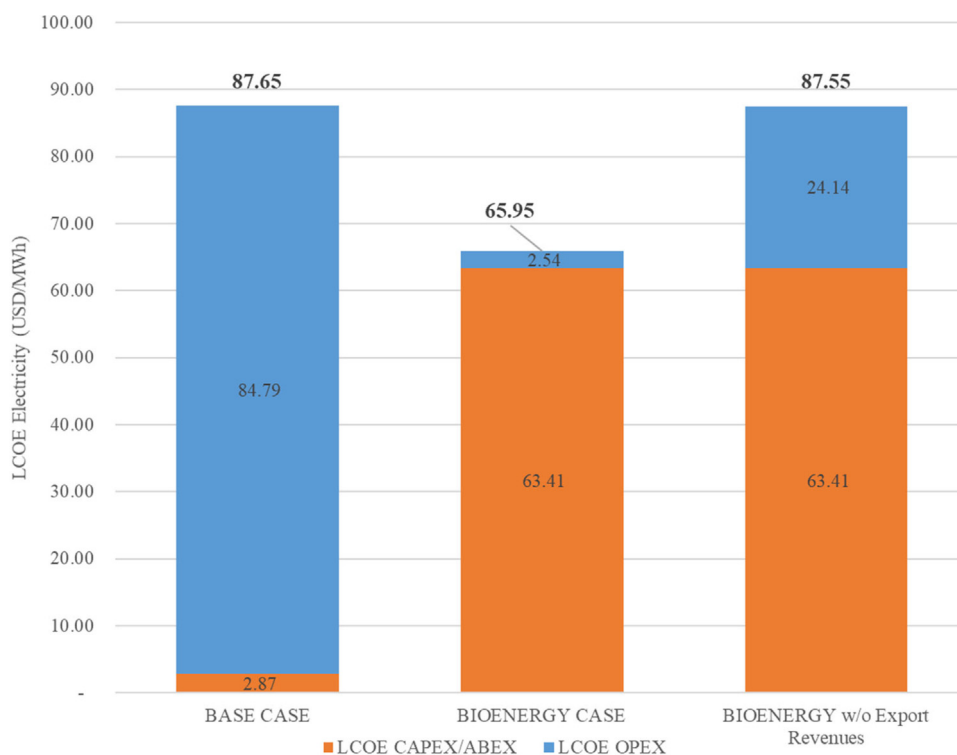


Fig. 8. LCOE comparison for electricity at TANWAT, with and without power sales to TANESCO.

combustion-based bioenergy at this enterprise. Even with a significant need for electricity, a minimum capacity factor of 79% is required to achieve a cost reduction in electricity, indicating that combustion-based CHP would not necessarily be economically attractive if electricity demand was the predominant requirement of the business.

This largely explains why there has been only limited adoption of bioenergy-based CHP production in the wood processing sector, with no new wood processing CHP projects commissioned

in the past five years. Only one other wood processing company (Mufindi Paper Mills) uses bioenergy-based CHP production and – together with TANWAT – also sells power to the grid under a legacy PPA with TANESCO. Other wood processors in Tanzania have insufficient internal requirements for heat and power to justify a CHP investment. Several have considered investing, with an eye on exporting surplus electricity to the grid, but have refrained from doing so as their internal heat and power demands are insufficient, new PPAs for sale of surplus electricity

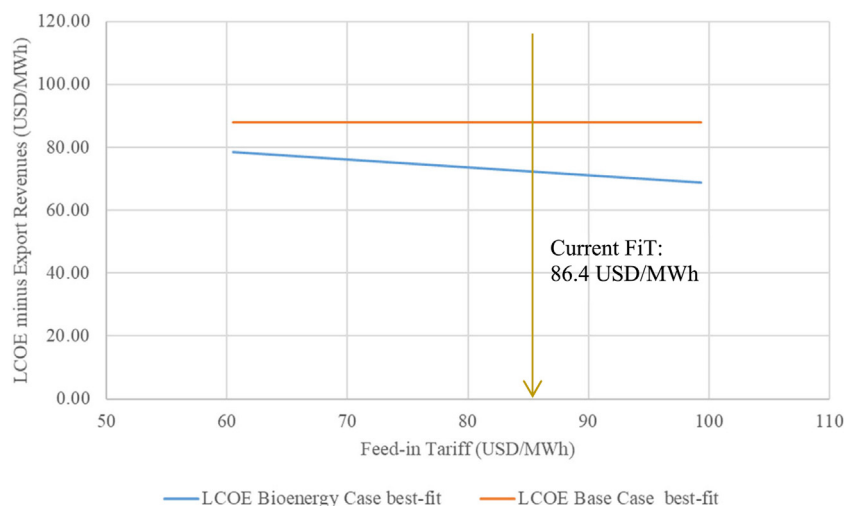


Fig. 9. Sensitivity of net electricity costs to feed-in tariff.

to TANESCO are not available, and they would only be eligible for commercially unviable feed-in tariffs.

In sum, while the wood processing industry in Tanzania generates considerable (and growing) quantities of residues, a combination of low internal heat and power demand at processing enterprises, relatively cheap and relatively reliable grid power, low FiTs and a government focus on large-scale hydropower, discourage new investments in bioenergy technology. Thus, while the TANWAT model has the potential to develop strong rural embedded generation, support local sustainable development objectives and reduce demand for fossil fuels for electricity generation to meet Tanzania's climate change targets, replication in today's political and economic environment is unlikely.

Wider use of wood residues for bioenergy in Tanzania's wood processing sector would require the energy regulator to increase its FiTs for wood bioenergy CHP generation (to ~\$0.20 per kWh), to reflect actual commercial costs, along with a commitment from TANESCO to enter into long-term PPAs that guarantee the purchase of electricity from new, or expanded, wood residue-based CHP projects.

3.4. Case study 3: Anaerobic digestion for heat production: Palm oil mill in Ghana

Agriculture dominates Ghana's economy and accounts for 54% of GDP, 40% of export income and 52% of labour force employment or engagement (FAO, 2020). The majority of farms are very small and only 15% are above 2 ha (ibid.). The main agricultural crops are roots and tubers, dominated by cassava. Tree crops such as cocoa, oil palm, coffee and rubber are dominant in the southwest.

Palm oil is Ghana's most important non-starch crop and edible oil. Ghana produces about 2.6 million t of oil palm fresh fruit bunches (FFB) per annum, grown on ~370,000 ha, from which 312,530 t of palm oil is extracted (FAO, 2020). 88% of the oil palm area is under smallholder management, which only accounts for about 60% of oil output as average productivity is lower than large estates with their own processing mills (Osei-Amponsah et al., 2012; Preferred by Nature, n.d., 2021).

The processing of FFB generates large quantities of solid residues, mainly empty fruit bunches (EFB), fruit fibre and palm kernel shell (PKS), as well as liquid residues in the form of palm oil mill effluent (POME).

Small-scale processors use manual techniques characterised by low extraction rates, but have little motivation to adopt modern bioenergy technology for heat or power production, due to insufficient feedstock beyond that required for process heat, lack of demand for additional energy, the standardisation of boilers designed for >10 t/hr processing capacity (with customisation for smaller sizes prohibitively expensive), lack of capital and a lack of skilled labour to operate and maintain more sophisticated systems.

It is therefore only the country's largest processors that have sufficient throughput, financial capacity, energy demand and human resources to justify investments in modern bioenergy technology. These are Ghana Oil Palm Development Company (GOPDC), SOCFINAF Ghana, Twifo Oil Palm Plantation, Norpalm Ghana, Benso Oil Palm Plantation and Juaben Oil Mills.

The six largest processors have combustion-based CHP plants at their mills that use solid oil palm residues (EFB, fruit fibre and PKS) for generating electricity for their own operations and administrative support functions, as well as low pressure steam to supply process heat for the various stages of oil extraction. The remainder of their power requirements are met through a combination of grid electricity and back-up diesel generators. Combustion-based CHP systems are a straightforward and economically viable solution for these larger mills to generate power – and especially heat – and therefore form the industry standard 'Base Case' for this analysis.

Palm oil processing also generates polluting liquid waste in the form of POME. Raw POME contains 90%–95% water and includes residual oil, soil particles and suspended solids. POME is acidic (pH 4 to 5) and has high BOD and COD of 25,000 and 50,000 mg/l, respectively (Kamyab et al., 2018), introducing a high risk of aquatic contamination if not properly treated and disposed of. The industry norm at Ghana's large mills is for POME to be discharged to aerobic treatment ponds for 90 to 120 days and then used for irrigation. This approach releases methane into the atmosphere and risks the leaching of contaminants into soil and groundwater.

Much research has been dedicated to alleviating the environmental threat posed by POME disposal, and AD is one solution that can reduce this threat, while simultaneously producing heat and/or power, plus benign liquid digestate with nutritional benefits for soils. GOPDC uses the POME as the primary fuel for an AD system to generate biogas to supply heat to its oil refining operation, which represents the Bioenergy Case for this demand sector. This research explores the commercial opportunity for

Table 4
Key economic modelling parameters for Case Study 3.

Category	Parameter	Value	Reference
General parameters	Discount rate	10%	(Central Intelligence Agency, 2017) Field visit, (NIRAS-LTS et al., 2021) –
	General price growth rate	4.0% (Consumer Price Index)	
	Energy price growth rate	0% p.a.	
	Electricity retail price	GHS 0.8436/kWh (\$162.43/MWh) (as reported by GOPDC)	
	Currency exchange rate	5.19 GHS/\$(3-yr average)	
Base Case	Fuel for combustion-based CHP	EFB, PKS, fruit fibre. LHV 19.45 GJ/t	Field visit, (NIRAS-LTS et al., 2021)
	Estimated cost of fuel for CHP	88% from own processing (nil cost) plus estimated 1,000 t/yr of PKS (~18% of fuel usage) purchased @ GHS 125/t	
	Fuel for refinery boiler	Transport diesel. LHV 36 MJ/l. Cost \$0.95/l	
	Diesel consumption for refinery boiler	1,692,361 l/yr	
	Nominal power	CHP unit: 1.5 MW _e , 12% electric eff., 68% thermal eff. Boilers (for CHP & refinery): ~14 MW _{th} , 85.5% eff.	
	Capacity factor	52% (4,520 operating hrs/yr)	
	Heat demand	14,470 MWh _{th} /yr (refinery, supplied from diesel boiler) 19,947 MWh _{th} /yr (mill, supplied from CHP unit)	
	Electricity sources	CHP plant 29.2% (3.52 GWh/yr) Grid 67.4% (8.11 GWh/yr) Gensets 3.4% (0.41 GWh/yr)	
Bioenergy Case	CAPEX	EUR 4.5 M (\$5.5M) for AD plant + additional biogas boiler	Field visit, (Gonzalez et al., 2014; NIRAS-LTS et al., 2021) Field visit, (NIRAS-LTS et al., 2021)
	OPEX reduction	Savings of 910,000 l/yr of diesel for 140,000 t of FFB processed, with slight increase in operating costs	
	Biogas yield	15.5 Nm ³ /m ³ of POME (55% methane)	
	POME disposal cost	\$0/m ³	
	POME treatment cost	\$0/m ³ (various values modelled later in sensitivity analysis)	
	Additional electricity demand (AD plant)	0.264 GWh/year (source: GOPDC Field visit)	

further adoption of POME-based AD for bioenergy in the palm oil industry in Ghana.

Table 4 summarises the data used in the LCC model for the GOPDC AD plant, as the Bioenergy Case, comparing it with a Base Case scenario of generating refinery heat using diesel-only boilers.

In this case study, the fixed costs of the CHP unit (CAPEX and fixed O&M costs) are allocated to LCOE_{heat} and LCOE_{electricity}, according to plant generation efficiencies (default option), since the CHP plant was operating in both the Base Case and the Bioenergy Case to cover both electricity and heat demands.

Applying these input parameters, the LCC model shows that the GOPDC plant has slightly lower LCOE for both electricity and heat under the Bioenergy Case than the Base Case (Fig. 10), in which the POME-fed AD plant has been installed to supplement the existing combustion-based CHP plant. A reduction of 6% is observed in the LCOE_{heat} (from \$64.6/MWh to \$60.6/MWh).

The value of the annualised CO₂ cost is 0.12 \$/t CO₂, which means the Bioenergy Case saves 3,371 t of CO₂ per US\$ million of investment and a total savings of 16,844 t of CO₂ over 10 years.

The introduction of the AD plant results in a change to the LCOE_{heat} breakdown, with higher contribution from CAPEX and lower share made up by OPEX. This would result in lower overall LCOE values if a longer evaluation period was considered, beyond the 10 years modelled. For example, Fig. 11 shows the impact of extending the period of analysis to 20 years, which produces a reduction in LCOE for heat of about 21% compared with the Base Case.

Savings on aerobic POME treatment and benefits from up-graded digestate for soil improvement were not considered in this economic analysis.

Adjustments were made to selected parameters in the LCC toolkit to determine their impact on LCOE, and the implications of

these changing costs for the replication potential of the Bioenergy Case.

For the introduction of AD in the palm oil sector, the key parameter investigated in the sensitivity analysis was the thermal energy demand for the refinery operations. Fig. 12 shows the linear regression line of 125 simulated scenarios, for both the Bioenergy Case and the Base Case. The current heat demand at the refinery is 14,470 MWh_{th} per year, which is only just above the tipping point of economic viability of this investment in an AD plant. For an operation with similar input specifications, heat demand below that level would not justify such an investment, while higher heat demand would result in lower LCOE values for the Bioenergy Case.

Another parameter considered was the price of diesel, as the consumption of diesel is an important component in the LCOE breakdown (Figs. 10 and 11), to supplement heat producing by burning biogas. The analysis in Fig. 13 shows that the Bioenergy Case is less sensitive to changes in diesel fuel price than the Base Case (as indicated by the shallower line gradient), since the quantity of diesel consumed is lower than it is in the Base Case. The LCC model shows that a drop in the cost of diesel to about \$0.71/l (from the current level of \$0.95/l) would outweigh the benefit of investing in AD, from a purely financial perspective.

Finally, the effect on LCOE_{heat} of POME treatment cost was investigated, for a range from \$0 to \$25/m³, based on comparable reference examples (Ahmad et al., 2009) (see Fig. 14). The results show that if the avoided cost of POME treatment is considered in the economic model, then the investment in an AD plant is shown to yield much more significant savings in the cost of heat supply than the basic analysis that considers only diesel savings.

The economic viability assessment indicates cost savings from AD by substituting biogas for diesel to generate heat for palm

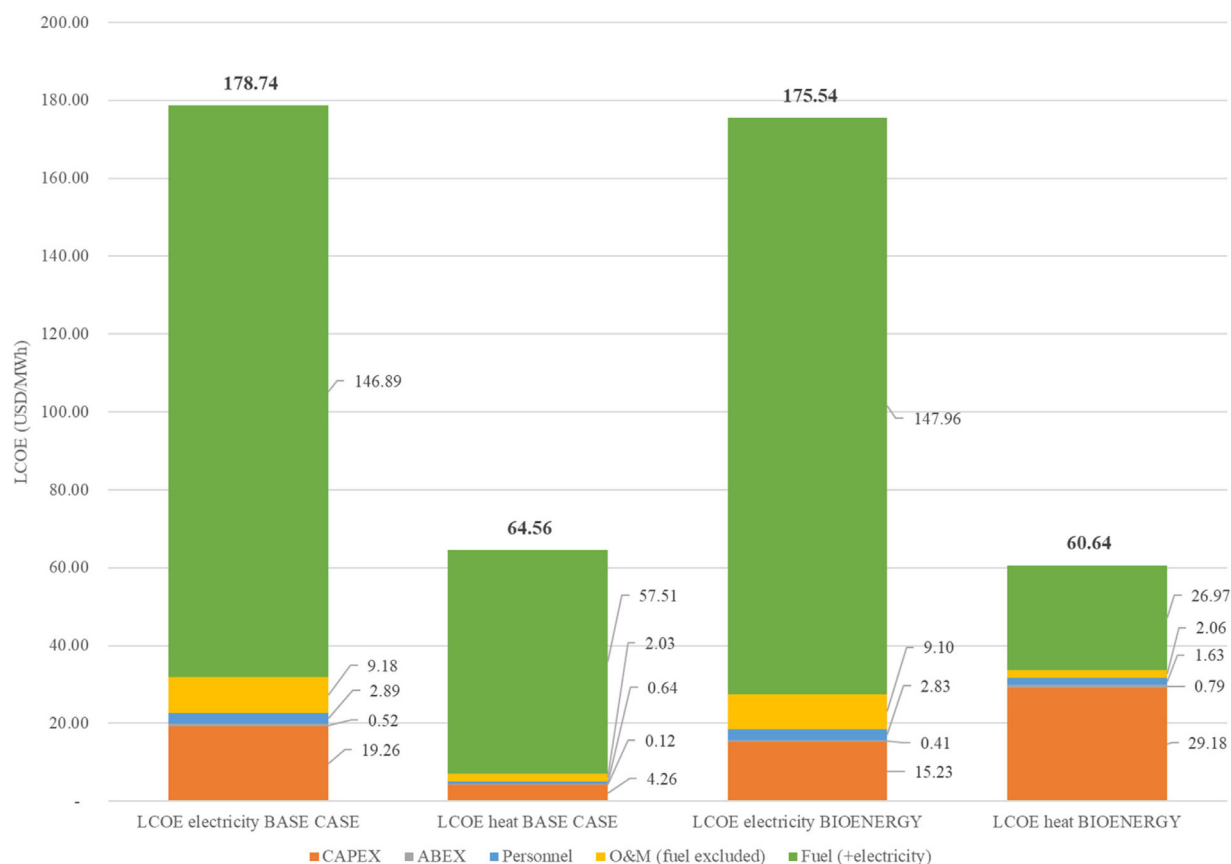


Fig. 10. LCOE comparison for power and heat, Base Case vs. Bioenergy Case.

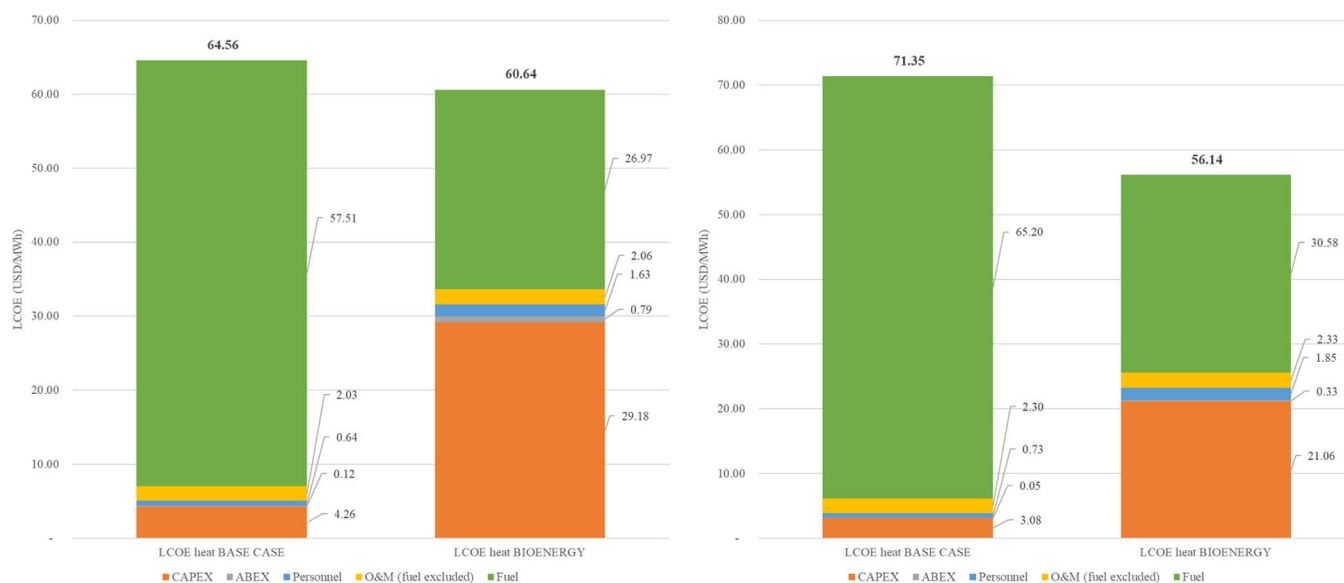


Fig. 11. LCOE comparison for heat, Base Case vs. Bioenergy Case over 10 years (L) and 20 years (R).

oil refining. The need for supplementary process heat was probably the main driver for GOPDC's investment in AD. In contrast with AD investments in other industries and in other countries, demand for electricity was not a motivating factor for the investment, given sufficient availability of power from the company's existing combustion-based CHP plant and the national grid.

From a commercial perspective, wider adoption of AD in Ghana's palm oil industry is hampered by the fact that all the

large processors already have combustion-based CHP plants that meet their requirements for process heat and contribute to their power needs, with the balance of electricity available relatively cheaply from the grid. Only one company (Juaben Oil Mills), in addition to GOPDC, has an oil refinery with its additional demand for process heat. However, Juaben's owners have decided to expand their combustion-based CHP capacity to meet this demand, rather than set up an AD plant, even though this will require them

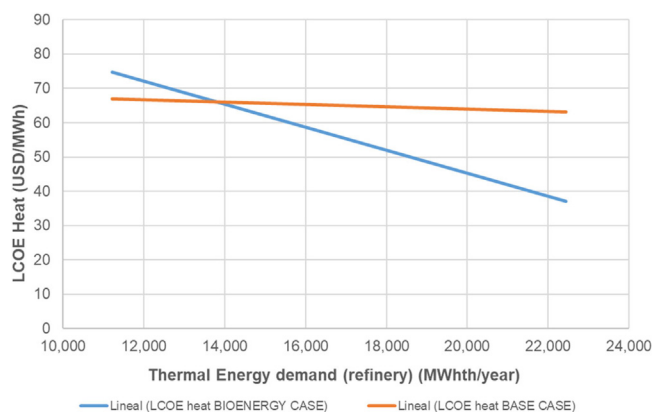


Fig. 12. Sensitivity of $LCOE_{heat}$ to total refinery heat demand.

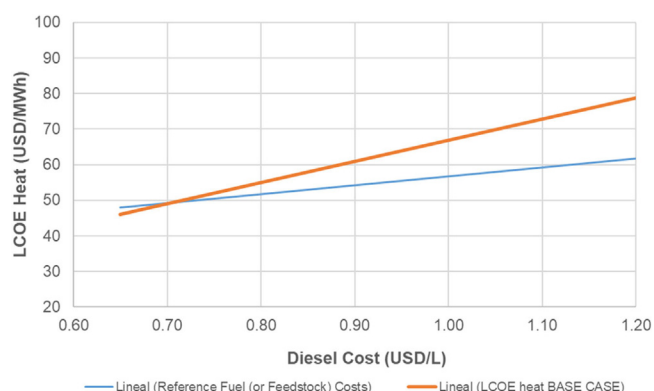


Fig. 13. Sensitivity of $LCOE_{heat}$ to diesel cost for boiler co-firing.

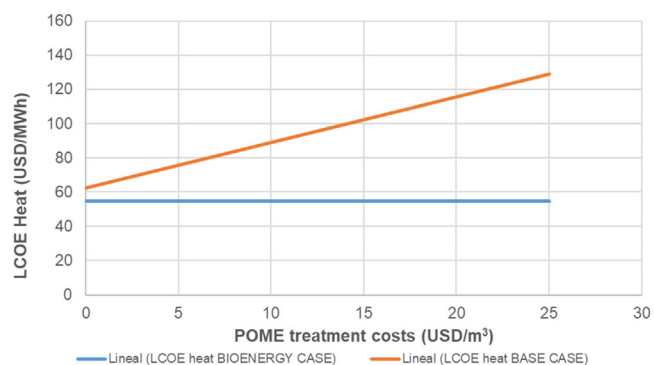


Fig. 14. Sensitivity of $LCOE_{heat}$ to POME treatment cost.

to buy in supplementary solid palm residues to provide sufficient fuel. Other motivations for adopting AD, such as more rigorous enforcement of POME treatment and disposal regulations, are unlikely to make the commercial case significantly stronger in this sector, given that there are less costly alternatives for effective POME management. While the efficacy of AD digestate as a fertiliser for oil palm plantations is not yet fully determined, this may improve the business case for AD in the future.

4. Conclusions and further research

4.1. Versatility of the LCC toolkit developed

The LCC modelling toolkit developed under the “Bioenergy for Sustainable Local Energy Services and Energy Access in Africa”

project (NIRAS-LTS et al., 2021) offers a user-friendly means of determining the economic feasibility of bioenergy investments in various (agro-)industrial sectors in Africa.

The toolkit has been designed in a versatile way for application in a variety of bioenergy pathways by a range of potential users, from a plant manager to a bioenergy project developer or a potential investor.

From the perspective of project assessment, the goal was to inform the investment planning needs of prospective developers, based on a multi-layered model structure. The first layer permits a rapid analysis based on parameters of energy consumption for different industrial sectors, predefined efficiencies for different categories and alarms when energy balances are not fulfilled. Values are pre-entered for CAPEX, OPEX and economic parameters for each country (consumer price growth rates, energy price growth rates, discount rates), though can be overwritten. The second layer permits more detailed technical definition of consumption, technical solutions and economic data. Both layers keep the structure of results, with clear and concise technical, economic and environmental KPIs. A module containing multi-parametric sensitivity analyses for key parameters is also incorporated. This layer could enable planners to develop appropriate subsidy schemes for technologies that could contribute to achieving a country's policy objectives for climate change mitigation and energy self-sufficiency.

In terms of technical versatility, the tool covers different bioenergy pathways, including AD and direct combustion, to be used for electricity production, heat production or both. This allows users to evaluate and compare different bioenergy cases, comparing them to a BAU base case in the chosen sectors.

The tool's main parameters were obtained from actual operational examples of the different technologies in use in SSA, continuously adding feedback from final users of existing bioenergy plants to incorporate actual values for CAPEX, O&M and other relevant costs. A thorough beta testing of the tool took place, both within the project team and with intended users.

However, the most important breakthrough of the toolkit does not concern any of its internal features, but is the combination of all the previously mentioned aspects, gathered in a common tool focusing on bioenergy for industrial processes for Africa, that can be customised and adapted as prices, legislation and technical aspects change, while maintaining the structural validity of the methodology.

4.2. Areas for further research

While the LCC toolkit incorporates important additions to previous models that add value for users, further features could bring out additional economic considerations and help foster the development of modern bioenergy pathways on the African continent.

The first would be to incorporate an economic cost-benefit of changes in net CO_2 emissions. The economic impacts of the carbon market, with a continuous growth trend due to the risks associated with climate change (Nordhaus, 2017), need to be more thoroughly analysed and incorporated, with various scenarios in the tool.

A second important area of further research would be the conversion of the Excel toolkit (designed as a DSS in this project) into a web-based tool that could incorporate real life cases and could allow continuous improvement (with updated databases on industrial energy consumption for different sectors and for successfully implemented projects). This could also lead to the development of different marketplaces that could foster the development of these technologies in the industrial sector in Africa.

CRedit authorship contribution statement

Oriol Gavalda: Conceptualization, methodology definition, calculations, analysis of results and discussion, conclusions and writing. **Arnau González:** Conceptualization, literature review, methodology definition, calculations, analysis of results and discussion, conclusions and writing – original draft preparation. **Mariana Raya:** Calculations. **Matthew Owen:** Field data collection, context characterisation, results and discussion. **Francis Kemausuor:** Field data collection, context characterisation, results and discussion. **Pol Arranz-Piera:** Conceptualization, methodology definition, context characterisation, analysis of results and discussion, conclusions and writing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The research presented in this paper was supported by the UK Foreign, Commonwealth and Development Office, under a two-year project entitled 'Bioenergy for Sustainable Local Energy Services and Energy Access in Africa - Phase 2' (BSEAA2), commissioned to a team led by NIRAS-LTS in partnership with Aston University and the consulting companies E4tech and AIGUASOL.

BSEAA2 was intended to support the development of innovative, commercial bioenergy pathways and technologies to accelerate the adoption of bioenergy in SSA, focusing on opportunities for AD and combustion for electricity and/or heat generation in the range 10 kW to 5 MW, with a Technology Readiness Level of 5+. The team investigated the challenges and opportunities affecting the commercial deployment of these technologies in Ethiopia, Ghana, Kenya, Mozambique, Nigeria, Rwanda, South Africa, Tanzania, Uganda and Zambia under six themes: biomass resources, technology, economics, business models, institutional, market and regulatory frameworks, and gender and inclusion.

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